

Han-Teng Liao

Socio-Technical System Architect • Ex Associate Professor and Vice Dean

Taipei and Penang

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Professional Service

Peer Reviewers, mostly Q1 and Q2 journals,

Editorial

Completed 20+ peer reviews, especially on AI and Industry 4.0. **Scientific Reports** (Q1:2) Financial Innovation (Q1:2); Nature **Humanities and Social Sciences Communications** (Q1:1); Politics and Governance (Q2:1), Frontiers in Public Health (Q1:1); Journal of Clinical Epidemiology (Q1:1); Heliyon (Q1:2); Cogent Economics & Finance (Q1:1); Cogent Education (Q2:1); Future Internet (Q2:1); **Discover Artificial Intelligence** (1); Sustainability (Q2:4); Applied Economics Letters (Q3:1); Environment Systems and Decisions (1);

2021 — Present

TC Member, Joint IEEE Systems Engineering/TEMS TC on Carbon Neutrality

Technical

Explore active research and strategies to promote carbon neutrality best practices, including surveying liaisons with other IEEE TCs, and other international organizations.

2026 — Present

Expert Contributor

International
Telecommunication
Union (ITU-T)

Focus Group on AI for Agriculture (FG-AI4A)

2023 - 2024

- Authored technical input document b-FG-AI4A-I-132 on linking carbon accounting data with agricultural business and products.
- Successfully integrated the framework into global standard ITU-T Y.Suppl.83 (07/2024) under Section 7.2.7 (TG-Digital Agri-Finance).
- Operationalized 'Carbon Information Chains' to translate regional standards into international functional requirements.

Student Branch Development & Launch

Guangzhou, China

IEEE Technology and Engineering Management Society (TEMS)

2019 - 2023

- Orchestrated and managed the establishment of the IEEE TEMS Student Branch during tenure as Center Director.

Education

DPhil (Ph.D.) in Information, Communication, and Social Sciences

Oxford, United Kingdom

University of Oxford

10/2007 - 10/2014

M.Sc. in Computer Science and Information Engineering

Taipei, Taiwan

National Taiwan University

01/08/2000 - 01/06/2004

M.S.S. (Master of Social Science) in Journalism

Taipei, Taiwan

National Taiwan University

01/08/2000 - 01/06/2004

B.Sc. in Electrical Engineering

Taipei, Taiwan

National Taiwan University (Includes a year-long exchange at the University of British Columbia)

01/10/1994 - 01/06/2000

Bachelor of Arts in Foreign Languages and Literatures

Taipei, Taiwan

National Taiwan University

01/08/1995 - 01/06/2000

Work Experience

Editor & Project Founder (Public Service Initiative)

Penang, Malaysia

scvcs.footprints.cv - SSbD Chips Value Chain Services| GBS.today - Geopolitical Business Shifts & Strategies

01/01/2026 - Current

Data Scientist and CEO

Guangzhou, China

Oxford Roadmapping	01/01/2023 - Current
Director, Higher Education Impact Assessment Center Nanfang College of Sun Yat-Sen University	Guangzhou, China 01/12/2019 - 01/01/2023
Associate Professor & Vice-Dean, Internet and New Media Nanfang College of Sun Yat-Sen University	Guangzhou, China 01/01/2017 - 01/01/2021
Research Fellow United Nations University (UNU) Institute on Computing and Society	Macau, China 01/07/2015 - 01/11/2016
Yahoo! Fellow, Institute for the Study of Diplomacy Georgetown University	Washington DC, United States 01/08/2010 - 01/07/2011
Doctoral Research Fellow Institute of Sociology, Academia Sinica	Taipei, Taiwan 01/08/2009 - 01/06/2010
Political Warfare Officer (Mandatory Service) Taiwan Military	Taiwan 01/11/2005 - 01/12/2006
Project Manager Open Source Software Foundry (OSSF), Institute of Information Science, Academia Sinica	Taipei, Taiwan 01/10/2004 - 30/11/2005
Action Researcher & Graduate Teaching Assistant Institute of Journalism, National Taiwan University	Taipei, Taiwan 01/09/2002 - 01/12/2003
Programmer & Independent Developer (Digital Publishing) Yuan-Liou Publishing Co., Ltd.	Taipei, Taiwan 01/07/2000 - 31/08/2000

Awards

Certificate of Merit Recognized for outstanding academic contributions	IEEE TEMSMET Conference 2021
Awarded for CSR (Corporate Social Responsibility) and ESG (Environmental, Social, and Governance) reporting research.	
Province-Level 'First Class' Course Curriculum Design & Pedagogy Excellence	Guangdong Department of Education 2020
Recognized for the development and instruction of the Capstone Project: 'API, machine learning and AI'.	
Nation-Level First Prize Instructional Excellence	National College Art & Design Awards 2020
Awarded for Best Teaching for Capstone Project: 'API, machine learning and AI'.	
OxTALENT Awards Best Interactive Infographics	University of Oxford 2014
Recognized excellence in university teaching using interactive digital content.	
Best Student Paper Award Research Excellence	Chinese Internet Research Conference 2009
Awarded for foundational digital community research.	
PGP Corporation and OII Scholarship Merit-based Doctoral Scholarship	Oxford Internet Institute, University of Oxford 2007

- Value: £12,000 to support DPhil research.

Taiwan Merit Scholarship

Doctoral Research Fellowship

Taiwan Government

2007

- Awarded for doctoral research in Digital Content (TWD \$2,000,000).

Publications

On the Electrodynamics of Moving Bodies

Annalen der Physik

September 1905

On the motion of small particles suspended in liquids at rest required by the molecular-kinetic theory of heat

Annalen der Physik

July 1905

Skills

“Strategic Architecture & Governance”

“AI System & Platform Architecture, ICT System Architecture & Analysis, Systemic Design Thinking, Software Specification Analysis, Architecture Generalization”

“Intelligence & Foresight (Data Science)”

“Strategic Foresight, Scenario Planning, Scientometrics, Bibliometrics, Science Mapping (Bibliometrix, VOSviewer), Webometrics, Knowledge Management”

“AI, Data Architecture & Automation”

“RAG (Retrieval-Augmented Generation) Context Engineering, Prompt Engineering, LLM-Driven Strategic Intelligence, Python (Pandas/NumPy), Semantic Data Modeling, Resource Description Framework (RDF), CMS Orchestration (Quarto, Obsidian, GitHub)”

“Product & Agile Leadership”

“Full-Lifecycle Product Management (MVP to Scale), PRD/MRD/BRD Documentation, UX/UI Design Research, Scrum, Agile Workflows, GitHub-Based Collaborative Engineering”

“Language Skills”

“Chinese (Mother tongue), English (Fluent — C2 across Listening, Reading, Spoken Production, Spoken Interaction, and Writing)”